CAS / Noetic Compass Technical Review Brief

Date: 2026-06-15
Status: working review draft
Purpose: request technical feedback, not announce a finished product

One-Sentence Summary

CAS / Noetic Compass is an attempt to turn model behavior and internal signals into structured notation, plain-English explanations, and reviewable evidence for human reviewers.

What It Is

The broader project has three connected parts:

- Noetic Compass: the assessment and explanation surface.
- CAS: the notation system for representing cognitive structure, tension, polarity, and interpretation patterns.
- CAS Harness: the audit layer that tests open-weight models, captures evidence, and packages findings for review.

In plain English: this is not only trying to see what a model says. It is trying to create a clearer record of how a model appears to move from prompt, to internal signal, to output, to reviewer-facing explanation.

Possible Contribution

The possible value is a layer between raw model-debugging data and human review.

CAS Harness is exploring whether model behavior and internal signals can be mapped into:

- CAS notation
- plain-English explanations
- reviewer flags
- saved Evidence Packets
- repeatable audit records

The intended safe review output is `FLAG_FOR_HUMAN_REVIEW`. In plain English, that means: "this deserves a person looking at it," not "this is approved."

Relationship To Interpretability Tools

This is not being presented as a replacement for mechanistic interpretability, sparse autoencoder work, activation steering, attention analysis, log-prob analysis, or internal model-debugging tools.

Those tools are stronger for low-level model-builder work. CAS Harness is asking a different question:

Can model behavior and internal signals be turned into a structured review record that a human can understand, challenge, preserve, and compare across runs?

Current Boundaries

This should be reviewed as an evidence and explanation layer, not as a finished safety certification system.

It is not currently claiming:

- customer-production `ACCEPT` verdicts
- frontier-scale validation above the tested scope
- prompt-injection prevention, removal, or reversal
- hallucination prevention
- general AI safety certification
- proof that CAS is uniquely superior to every matched alternative

Those items are still `OPEN` / `NOT LICENSED` unless a specific test closes them.

What Exists Now

Current work includes a public Noetic Compass platform concept, a formal CAS notation framework, compiler-grounded CAS / CAS Native research, CAS Harness experiments on open-weight models, and Evidence Packet / WORM preservation work in the production path.

The areas that most need stronger proof are:

- whether CAS adds information beyond good matched controls
- whether CAS-specific notation is better than alternative taxonomies
- whether internal signals are strong enough for reliable audit use
- whether the product path is clean enough for outside users
- whether reviewers actually find the explanations useful

What I Want Feedback On

I am looking for grounded technical critique before I overbuild, overclaim, or send the wrong material to the wrong audience.

The questions I most want answered are:

1. Is this meaningfully distinct from existing evaluation, debugging, and interpretability workflows?
2. Where are the claims too weak, too broad, or too early?
3. What evidence would you want before taking this seriously?
4. What should be removed, simplified, or tested next?